Table 11.4: Observed frequency for different situations

S.No.	Situation	Observed frequency
1	L moves toward the stationary S	$f' = \left(\frac{v + v_L}{v}\right) f$
2	L moves away from the stationary S	$f' = \left(\frac{v - v_L}{v}\right) f$
3	S moves toward the stationary L	$f' = \left(\frac{v}{v - v_S}\right) f$
4	S moves away from the stationary L	$f' = \left(\frac{v}{v + v_S}\right) f$
5	S and L move toward each other	$f' = \left(\frac{v + v_L}{v - v_S}\right) f$
6	S and L recede from each other	$f' = \left(\frac{v - v_L}{v + v_S}\right) f$
7	S chases the L	$f' = \left(\frac{v - v_L}{v - v_S}\right) f$
8	L chases the S	$f' = \left(\frac{v + v_L}{v + v_S}\right) f$
9	S and L move toward each other and the medium also moves in the direction of sound with speed v_m	$f' = \left(\frac{(v + v_m) + v_L}{(v + v_m) - v_S}\right) f$

Note

Doppler effect in sound is asymmetric while that in light is symmetric.

The observed frequency of sound when the source moves toward stationary listener and the observed frequency when the listener moves toward stationary source with the same speed are not equal. Although the relative speed is same in both the cases, observed frequency is different. Hence, we say that the Doppler effect in sound is asymmetric. The reason is that sound wave requires a medium for its propagation and it has its speed with respect to that medium.

But in the case of light and other electromagnetic radiations, the observed frequency is the same in both abovesaid cases. Therefore, Doppler effect in light and other electromagnetic waves is symmetric because the propagation of light is independent of the medium.

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